

Wenge Rao

Tel: (+86) 18785012030, E-mail: wgrao@stu.pku.edu.cn

Homepage: giselle1119.github.io

EDUCATION

Ph.D. in Physical Geography Peking University, Beijing	Sept. 2020 - June 2026 (expected)
Joint Ph.D. in Public Administration Baruch College CUNY, New York	Dec. 2024 - July 2025
B.S. in Ecology Guizhou University, Guizhou	Sept. 2016 - June 2020

PUBLICATION

Peer-Reviewed Publications

1. Rao, W., Shen, Z., & Duan, X. (2023). Spatiotemporal patterns and drivers of soil erosion in Yunnan, Southwest China: RUSLE assessments for recent 30 years and future predictions based on CMIP6. *Catena*, 220, 106703. [Link]
2. Zhao, F., Yang, T., Luo, C., Rao, W., Yang, G., Li, G., & Shen, Z. (2022). Comparing elevational patterns of taxonomic, phylogenetic, and functional diversity of woody plants reveal the asymmetry of community assembly mechanisms on a mountain in the Hengduan Mountains region. *Frontiers in Ecology and Evolution*, 10. [Link]
3. Xie, Y., Shen, Z., Camarero, J. J., Peñuelas, J., Wang, X., Li, J., Rao, W., Chen, X., Zhao, F., & Xiao, F. (2025). Keys to the global treeline formation: Thermal limit for its position and moisture for taxon-specific variation. *Proceedings of the National Academy of Sciences (PNAS)*. [Link]

Preprints

1. Ning, C.#, Rao, W.#, Shen, Z., & Li, J. (2024). Hydrothermal gradient drives the elevational transition of the relationship between tree growth and forest canopy productivity. Preprint.

Manuscripts Submitted

1. Rao, W., Shen, Z., & He, G. (2026). Multi-objective land-use optimization incorporating renewable energy potential under climate change. Manuscript submitted.
2. Rao, W., Shen, Z., Liu, X., & Liu, Q. (2026). Drought vulnerability of fragmented forests in the Great Green Wall of China under climate change and human activities. Manuscript submitted.
3. Bao, J., Rao, W., Shen, Z., & Zhao, F. (2026). Species diversity patterns and assembly rules of elevational vegetation zones in Yunlong Tianchi, Hengduan Mountains: Integrating phylogeny and functional traits. Manuscript submitted.

Indicates equal contribution and shared first authorship.

AWARDS

- Xingye Scholarship, Peking University, 2023
- Research Excellence Award, Peking University, 2023
- Outstanding Presentation Award, The 19th Peking University Eco-Youth Forum, Dec. 2022
- Individual Innovation Award, Guizhou University, Dec. 2019
- Third Prize, Esri China GIS Software Development Competition, Geographic Design Track, Nov. 2019
- Excellence Award, Esri China GIS Software Development Competition, Remote Sensing Applications Track (as team leader), Nov. 2018

TALKS

- IPWSD (Interdisciplinary PhD Workshop in Sustainable Development at Columbia University) 2025, [*Multi-objective land-use optimization incorporating renewable energy potential under climate change*] , New York, Apr. 2025
- AGU24, [*Multi-objective land-use optimization incorporating renewable energy potential under climate change*] , Washington, Dec. 2024
- The IALE 2023 World Congress, [*Drought vulnerability of fragmented forests in the Great Green Wall of China under climate change and human activities*] , Kenya, July 2023
- The 11th IALE-China, [*Drought vulnerability of fragmented forests in the Great Green Wall of China under climate change and human activities*] , Beijing, June 2023
- The 19th Peking University Eco-Youth Forum, [*Spatiotemporal patterns and drivers of soil erosion in Yunnan, Southwest China: RUSLE assessments for recent 30 years and future predictions based on CMIP6*] , Beijing, Dec. 2022
- The 21th Conference on Ecology, [*Spatiotemporal patterns and drivers of soil erosion in Yunnan, Southwest China: RUSLE assessments for recent 30 years and future predictions based on CMIP6*] , Guizhou, Oct. 2022
- The 20th Conference on Ecology, [*Poster: Spatiotemporal patterns and drivers of soil erosion in Yunnan, Southwest China: RUSLE assessments for recent 30 years and future predictions based on CMIP6*] , Shanghai, Oct. 2021

SKILLS & INTERESTS

Skills: Proficient in R, Google Earth Engine, ArcGIS, QGIS, MATLAB, and ENVI

Interests: Enjoys ball sports, photography, calligraphy, and singing

SCHOOL ACTIVITIES

- Member, BNU Summer School on Geography and Sustainability 2024
- Chair, Sports and Culture Department, Graduate Student Association, Peking University, June 2021 - June 2022
- Goalkeeper, College Football Team & Point Guard, College Basketball Team, Peking University, Sept. 2020 - Present
- Assistant, Student Affairs Office, Peking University, Sept. 2020 - Present
- Member, School TV Department, Guizhou University, Sept. 2016 - June 2017